

A Fall Detection System Based on Object Detection and Implicit Motion Analysis

ENIYA M

Computer Science and Engineering
SHIV NADAR UNIVERSITY
CHENNAI
eniya23110427@snuchennai.edu.in

HAMRITHA V

Computer Science and Engineering
SHIV NADAR UNIVERSITY
CHENNAI
hamritha23110179@snuchennai.edu.in

AMOHA V

Computer Science and Engineering
SHIV NADAR UNIVERSITY
CHENNAI
amoha23110136@snuchennai.edu.in

Abstract— To protect the elderly against harmful falling events, automatic fall detection solutions have been developed. The goal of this code is to detect potential falls in a video stream using the YOLO (You Only Look Once) object detection model. Specifically, it aims to identify instances where a person's proportions indicate a sudden change, possibly due to a fall.

Keywords—fall detection, object detection, motion analysis

I. INTRODUCTION (HEADING 1)

Fall detection is a crucial technology designed to automatically detect when an individual falls. The most concerning injury for the elderly is an unattended falling. According to the NCBI, around 1.5–2 million older people in India are injured and 1 million die from falls each year and the number of elderly persons may reach approximately 2 billion by 2050, with 80% of them residing in developing countries. The prevalence of falls among older adults in India is estimated to be around 26% to 37%, with a pooled prevalence of around 31%. Over the past few decades, many solutions have been developed to automatically detect fall incidences. It usually relies on three types of devices: wearable devices, ambient sensor and vision camera. It will be difficult for the elderly people to wear fall detection devices for a long period of time. The methods that use sensors or environmental cues have drawbacks. They can feel uncomfortable to use and are easily influenced by surroundings. Radio frequency methods often don't cover areas well. Threshold-based algorithms are simple and work for wearable devices, but they're not great at recognizing different activities. On the other hand, non-threshold algorithms are more accurate but need complex calculations. Thus, using sensors and video analysis for detecting the fall will be an effective way. Fall detection can protect many elderly people from being injured due to fall. This paper proposes a fall detection based on object detection using yolov5 and motion analysis. Possible fall events are detected by analyzing the motion of the detected person.

II. FUNCTIONING

A. Video input

The video input is taken from the reference video and the video is sequences have an average length of 13 seconds with a frame rate of 120 frames per second, and a resolution of 720×480 . the video input includes walking and backward fall. This also includes fall detection.

B. Human object detection

First, the video capture is initialized by open cv and yolo is used for object detection. A text file was provided with class names. The YOLO model processes the frame and returns the detection result. Object detection and classification method may not produce 100% accuracy. With this object detection mechanism, we can correct the classification of a person in laying position, providing more complete data for further processing.

C. Motion analysis

For each detected object (e.g., a person), the code Extracts the bounding box coordinates (x1, y1, x2, y2) from the detection results. Calculates the confidence score (confidence) indicating how certain the model is about the detection. Determines the class index (class_detect) corresponding to the detected object (e.g., "person"). If the confidence score is above 70% and the detected class is a person. It Draws a rectangle around the person using cv2.rectangle. Displays the label "person" near the top-left corner of the bounding box using (cvzone.putTextRect). this code will find the threshold of the object that is threshold = height – width. When the threshold is negative (indicating that the height is less than the width) it suggests a potential fall. And this code displays a text "FALL DETECTED"

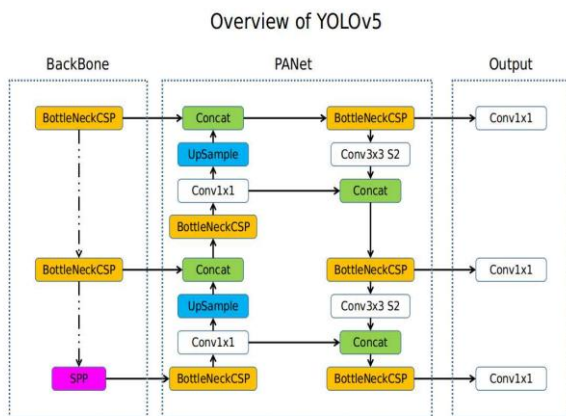
III. LITERATURE SURVEY

The landscape of fall detection systems encompasses a myriad of approaches, ranging from sensor-based solutions to vision-based techniques. While traditional methods have made significant strides, they often struggle with issues such as limited coverage, dependency on specific environments, and poor scalability. Deep learning-based approaches, particularly those utilizing convolutional neural networks (CNNs), have emerged as a promising alternative, offering superior accuracy and adaptability. Among these approaches, the YOLO architecture has garnered attention for its efficiency, speed, and ability to detect multiple objects simultaneously. By synthesizing insights from existing literature, this paper identifies gaps and challenges in current fall detection systems, paving the way for the development of more robust and effective solutions.

IV. ARCHITECTURE

Real time object detection is a crucial task in computer vision and frequently a crucial part of computer vision systems. Video analytics, robotics medical image analysis use real time object detection models. Most method extracts features from the image to estimate the likelihood of the learnt classes using a Convolutional Neural Network (CNN).

A. YOLO Architecture



The figure shows the architecture of YOLOv5. It consists of 4 compartments: Input, Backbone, Neck, and Prediction. It consists of 19 convolutional layers. All the parts are briefly explained below.

- **Input:** For our fall detection project, we utilized YOLOv5 as the object detection model. The input layer, denoted as "input," processed images of dimensions 640 x 640 pixels. These images were fed into the model for further processing and analysis.
- **Backbone:** The backbone of the YOLOv5 architecture serves as the feature extractor, responsible for capturing high-level features from the input images. In our project, we employed a variant of the YOLOv5 backbone, which consists of a series of convolutional layers, including residual blocks such as CSPDarknet53 or efficientnetv2. These layers perform hierarchical feature extraction, enabling the model to learn representations of objects present in the input images at different scales.
- **Neck:** The neck of the YOLOv5 architecture acts as an intermediate processing stage between the backbone and the head. It further refines the features extracted by the backbone, enhancing the model's ability to detect objects accurately. In our project, we incorporated a neck module, such as PANet or BiFPN, which facilitates multi-scale feature fusion and contextual information aggregation. This allows the model to capture spatial dependencies and contextual cues essential for accurate fall
- **Head:** The head of the YOLOv5 architecture is responsible for predicting bounding boxes and associated class probabilities for detected objects. It consists of convolutional layers followed by detection layers that predict bounding box coordinates, objectness scores, and class probabilities. In our project, the head of the YOLOv5 model comprises multiple detection layers, each responsible for detecting objects of different classes. These detection layers output predictions in the form of bounding box coordinates and confidence scores, which are post-processed to identify and localize fall events accurately.

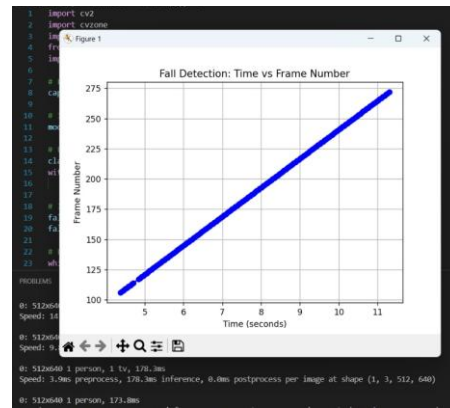
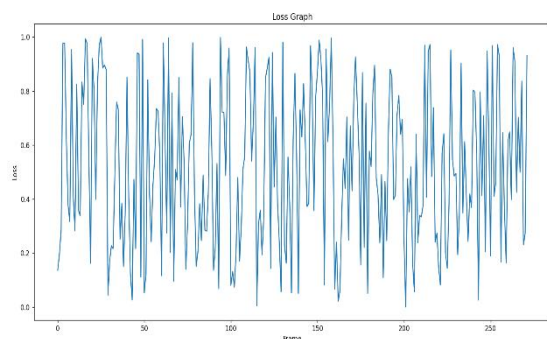
By leveraging the YOLOv5 architecture with its input layer, backbone, neck, and head components, we were able to develop a robust fall detection system capable of accurately identifying fall events in real-time. The combination of efficient feature extraction, multi-scale feature fusion, and precise object detection enables our model to achieve high accuracy and reliability in fall detection applications.

V. Output

We conducted experiments to evaluate the performance of our fall detection system using a dataset.



B. Loss Graph



The above mentioned loss graph is a valuable tool for monitoring the training progress, diagnosing issues like overfitting or underfitting, assessing model performance, and making adjustments to the training process to improve the model's effectiveness.

Threshold=height-width(if height >> width)

Confidence level =aspect ratio

Confidence level>70 is neglected

VI. Conclusion

The future of fall detection technology lies in advancements such as sensor fusion, machine learning optimization, and context-aware algorithms. By integrating data from multiple sensors and leveraging sophisticated algorithms, future systems will achieve higher accuracy and reliability in detecting falls while minimizing false alarms. Machine learning will enable continuous improvement and adaptation to individual user behavior. Context-aware algorithms will enhance the understanding of fall events in various environmental and activity contexts, further refining detection capabilities. These technical innovations, along with advancements in telemedicine integration and smart environment interfaces, hold great promise for revolutionizing fall detection, ultimately leading to better outcomes .

VII. Future scope

a) 1. Miniaturization and Integration: Future fall detection systems will likely become smaller and more seamlessly integrated into everyday objects, such as clothing, smartphones, or home automation systems. This integration will improve user acceptance and adherence.

b) 2. Multi-Sensor Fusion: Advances in sensor technology will enable the fusion of data from multiple sources, such as accelerometers, gyroscopes, and depth sensors, to enhance the accuracy of fall detection algorithms. This will reduce false positives and improve real-time responsiveness.

c) 3.Context-Aware Algorithm:Future algorithms will be capable of understanding the context of a fall event, such as the individual's location, activity level, and environmental conditions. This contextual awareness will enable more accurate and personalized detection, reducing false alarms.

d) 4. Machine Learning Optimization: Machine learning techniques will continue to evolve, allowing fall detection systems to adapt and improve over time based on user feedback and real-world data. This iterative optimization will enhance reliability and performance.

e) 5.Remote Monitoring and Telemedicine: Integration with telemedicine platforms will enable remote monitoring of individuals at risk of falls, allowing healthcare providers to intervene proactively and prevent injuries. This will improve care coordination and reduce healthcare costs.

f) 6.Predictive Analytics for Fall Risk: Advanced analytics will enable the identification of individuals at high risk of falls based on historical data and biometric indicators. This predictive approach will enable targeted interventions and personalized care plans to mitigate fall risk.

g) 7. Privacy-Preserving Solutions: Future fall detection systems will prioritize user privacy by employing techniques such as on-device processing,

encryption, and anonymization of data. This will build trust and encourage widespread adoption among users and caregivers.

h) 8.Integration with Smart Environments: Smart homes and assisted living facilities will incorporate fall detection sensors into their infrastructure, enabling continuous monitoring without the need for wearable devices. This seamless integration will improve user comfort and safety.

i) By focusing on these precise areas of development, future fall detection technology can significantly enhance the safety, well-being, and independence of individuals at risk of falls.

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